SAR Nonsparse Scene Reconstruction Network via Image Feature Representation Learning

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***Abstract*— Synthetic aperture radar (SAR) is widely used in various fields due to its all-weather and all-day working characteristics. With the increasing use of SAR on small platforms, SAR is facing a series of problems due to the large volume of echo data. Imaging methods based on compressed sensing (CS) use the sparsity prior of the scene to reconstruct images from undersampled echoes. However, the CS-based method requires the imaging scene or its transformation domain to be sparse, which is not the case for most practical applications. This article proposes a deep unrolling network named NSR-NET, which is based on SAR image representation learning and is applicable for undersampled imaging in nonsparse scenes. In modeling, the learned image representation is adopted as the regularization term. Then, the proximal gradient descent (PGD) algorithm was used to derive the iterative solution of the model. In network design, the iterative process is unrolled into a deep neural network with learnable parameters. Specifically, image representation is obtained through 2-D convolutional layers in the network, and a learnable piecewise linear layer is used to fit the regularization function, which ultimately achieves the mapping from undersampled echoes to SAR images. Comparative experiment using different imaging methods shows that the imaging performance of the proposed network exceeds that of the state-of-the-art methods in nonsparse scenes. Moreover, we also designed transferability validation experiments with different radar parameters and imaging scenes, whose experimental results suggest that the proposed network has good generalization ability.**

***Index Terms*— Deep unrolling network, image representation learning, nonsparse scene, proximal gradient descent (PGD), synthetic aperture radar (SAR), undersampling.**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active remote sensing system that uses microwave frequencies to achieve high-resolution imaging of the Earth's surface. Compared with passive optical imaging systems, SAR has advantages such as all-weather, all-day, and penetrating capabilities and has been widely used in tasks such as environmental monitoring, disaster warning, and urban mapping [1], [2], [3], [4]. In recent years, with the rapid expansion of application needs, the SAR imaging bandwidth and resolution have been continuously increased [5], [6], and the amount of echo data required for imaging has also increased. On the other hand, more and more SAR systems are placed on small platforms such as small satellites and unmanned aerial vehicles, which face great limitations in data storage and transmission. To fully use the advantages of small SAR platforms in applications, there is an urgent need for high-precision SAR imaging methods that can solve the conflict between high imaging resolution and low echo data volume.

The development of compressed sensing (CS) theory enables high-precision reconstruction of the original signal from compressed sampling observation signals under the premise that the original signal satisfies certain compressible conditions, which solves the above-mentioned problems in SAR imaging [7], [8], [9], [10]. In [11] and [12], the authors adopted imaging methods based on CS. Specifically, they established an approximate mapping between the scene and the echo and modeled the undersampled SAR echo imaging problem as an optimization problem with sparse constraints. Then, they used iterative algorithms of CS to reconstruct the sparse imaging scene. These methods only need significantly reduced amounts of echo data and alleviate problems such as SAR data storage and transmission. However, traditional iterative reconstruction methods based on CS for SAR imaging not only have long convergence times but also have difficult-to-determine iteration parameters. Furthermore, these methods require the imaging scene itself or its transformation domain to have a certain degree of sparsity. In reality, most observed scenes by SAR are not sparse, and it is also difficult to ensure sparsity under specific transformations.

The matrix sensing (MS) theory provides a new approach to solving imaging problems in nonsparse scenes [13], [14]. In [15], the authors have introduced the MS theory into the task of undersampled imaging. Furthermore, some scholars have modeled the undersampling imaging problem as a joint sparse and low-rank recovery problem, achieving good results in nonsparse imaging scenes [16], [17], [18]. However, both the low-rank-based methods and joint low-rank and sparse imaging methods face problems of long iterative time and manual parameter tuning. Moreover, these methods also assume that SAR images have certain properties, such as low rank or sparsity, which may not be reasonable in actual imaging tasks.

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Deep learning (DL) has been a hot topic in recent years, and introducing DL theory into SAR imaging can solve the problems mentioned above of long iterative time and manual parameter tuning. Mason et al. [19], Yonel et al. [20], and Hu et al. [21] have applied DL methods to SAR imaging problems and achieved ideal results. However, deep neural networks do not have interpretability, which is crucial in the SAR field [22].

The deep unrolling network was first proposed by Gregor and LeCun in [23], which possesses interpretability and parameter efficiency. Since then, the deep unrolling network method has been widely applied to various signal and image processing problems, including but not limited to optical image super-resolution [24], deblurring [25], [26], medical imaging [27], [28], [29], and speech signal processing [30], among others. These research works show that deep unrolling networks are widely used and perform well in various signal and image processing problems, demonstrating the effectiveness and superiority of the deep unrolling network method. Moreover, there are a large number of well-focused remote sensing SAR images publicly available, such as those from the MSTAR dataset [31] or Airbus sample imagery, which can be used as SAR image data in the SAR imaging dataset for training and testing deep unrolling networks.

Given the outstanding performance of deep unrolling networks and the ease of constructing SAR datasets, researchers have already applied them to tasks such as SAR/ISAR imaging, SAR super-resolution, and 3-D SAR sparse imaging [32], [33], [34], [35], [36], [37], [38]. In particular, in the task of reconstructing SAR images based on undersampled echoes, the use of unrolling networks has significantly improved imaging performance. In [38], the method unrolled the classical iterative shrinkage thresholding algorithm (ISTA) into a deep neural network, which achieved performance improvements compared with the iterative algorithm itself. However, its applicability was still limited by the sparse prior constraints in the modeling. To address this limitation, the state-of-the-art imaging methods based on unrolling networks have improved the reconstruction models for nonsparse scenes. An et al. [39] model the SAR image as a joint low-rank and sparse matrix. Then the iterative solution obtained from the alternating direction method of multipliers (ADMM) was unrolled into a deep neural network, achieving excellent reconstruction performance for nonsparse scenes. Huo et al. [40] used a learnable transformation operator to transform the complex-valued scene matrix, enforcing sparsity in the transformed domain and enabling the reconstruction of nonsparse scenes. It is found that there are two main approaches to solving the imaging problem of nonsparse images. The first one is to add prior constraints other than sparsity, such as low-rank characteristics, while the second approach is to transform the image through learnable operators or networks to make it satisfy the assumed prior conditions in the transform domain. However, these methods may lose important features of the image except for the assumed prior during imaging, leading to a decrease in imaging performance.

To overcome the limitations mentioned above, this article proposes an unrolling network for undersampled SAR imaging that uses image representation and validates the effectiveness of the method through simulation experiments. The main contributions of this article are as follows.

1) In the modeling of undersampled SAR imaging problems, we improve the conventional reconstruction model with prior constraints assumed in advance by designing an adjustable parameter model with image representation constraints for nonsparse scenes.
2) In the solution of the imaging problem, we propose a nonsparse scene reconstruction network named NSR-NET. Specifically, we approximate the iterative solution obtained by proximal gradient descent (PGD) and unroll it into a deep neural network with learnable parameters, where each iteration of the algorithm becomes a layer of the neural network. This approach enables the network to learn the image representation from the existing datasets.
3) We conduct multiple experiments to validate the effectiveness of the proposed method. These include the comparative experiment with the state-of-the-art imaging methods to demonstrate the efficiency and accuracy of the proposed method, transferability validation experiments with different radar parameters and imaging scenes to demonstrate the good generalization ability of NSR-NET, and so on.

The structure of this article is arranged as follows. Section II establishes the undersampled echo model; Section III proposes the undersampled SAR imaging model with image representation constraint and provides an iterative solution; Section IV presents an approximate iterative solution and proposes an imaging method based on the deep unrolling network; Section V shows the results and discussions of the conducted simulation experiments; and Section VI gives the conclusion.

II. Undersampled Echo Model

Assuming the radar platform moves parallel to the azimuth direction with a flight speed of v and the imaging scene of SAR is rectangular. To simplify the computation, the imaging scene can be discretized. That is, the imaging scene is composed of a series of grid-like distributed point targets, and the scattering coefficient of each point target describes the overall scattering characteristics in its surrounding small area. The distance between adjacent point targets is the imaging resolution, as shown in Fig. 1, where R_0 is the minimum slant range from the radar to the baseline, R_B is the minimum slant range from the radar to the position (m, n), M is the number of point targets (i.e., resolution cells) in the azimuth direction after discretization, and N is the number of distancewise discretization resolution cells. A certain element $S(p, q)$ in the discrete echo matrix can be expressed by the following equation:

$$S(p,q)=\sum_{m=1}^{M}\sum_{n=1}^{N}\sigma(m,n)w_r\big(\tau_p-\tau_q(m,n)\big)w_a(t_q)\cdot\exp\Big\{j\pi\gamma\big[\tau_p-\tau_q(m,n)\big]^2-j2\pi f_c\tau_q(m,n)\Big\} \tag{1}$$

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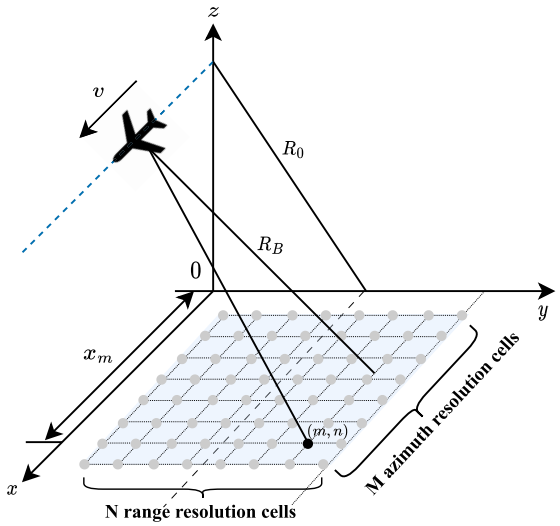

Fig. 1. Discretized observation geometry of SAR imaging.

where $p = 1, 2, \ldots, P$ represents the fast-time sampling points, $q = 1, 2, \ldots, Q$ represents the slow-time recording points; $\sigma(m, n)$ represents the scattering coefficient of the resolution cell located at (m, n); $w_r(\tau_p)$ and $w_a(t_q)$ represent the range pulse envelope and the azimuth window function, respectively; γ is the frequency modulation rate, and f_c is the carrier frequency; and $\tau_q(m, n)$ represents the radar echo delay of the resolution cell located at (m, n), which can be expressed as follows:

$$\tau_q(m,n) = \frac{2R(t_q; R_{B,n}, x_m)}{c} = \frac{2}{c}\sqrt{R_{B,n}^2 + (vt_q - x_m)^2} \quad (2)$$

where c is the speed of light, and $R_{B,n}$ and x_m represent the slant range and azimuth position of the resolution cell located at (m, n), respectively.

If we define $\mathbf{y} = \mathrm{vec}(\mathbf{S})$ and $\mathbf{x} = \mathrm{vec}(\mathbf{X})$, where $\mathbf{S} \in \mathbb{C}^{P\times Q}$ is the echo matrix with its (p, q) element being $S(p, q)$, and $\mathbf{X} \in \mathbb{C}^{M\times N}$ is the scattering coefficient matrix with its (m, n) element being $\sigma(m, n)$, then (1) can be expressed in the following form as shown in the following equation:

$$\mathbf{y} = \Phi\mathbf{x} \quad (3)$$

where $\Phi \in \mathbb{C}^{PQ\times MN}$ is referred to as the echo observation matrix, which has the form in (4), as shown at the bottom of the page. It can be easily derived from (4) that (1) can be simply expressed as $\mathbf{S}(p, q) = \phi(p, q)\mathbf{x}$. Equation (3) represents a 1-D SAR linear observation model. The process of SAR imaging can be regarded as solving the linear inverse problem of reconstructing the image vector $\mathbf{x}$ from the known echo vector $\mathbf{y}$, using either the echo observation matrix Φ itself or some of its information.

To reduce the amount of echo data, compressed sensing theory is used to perform sensing observation and reconstruction on the scene. When constructing a random observation matrix, since the known information that can be used to reconstruct the SAR imaging scene image is the echo data, it is necessary to design based on the SAR echo observation matrix Φ given in (4). One simple and effective design approach is to randomly select I rows ($I < PQ$) from Φ according to a certain rule to construct a random observation matrix $\Phi_{\mathrm{CS}} \in \mathbb{C}^{I\times MN}$, as shown in the following equation:

$$\Phi_{\mathrm{CS}} = \Xi\Phi \quad (5)$$

where $\Xi \in \mathbb{R}^{I\times PQ}$ represents the random undersampling matrix.

According to (5), if we multiply both sides of the 1-D SAR linear observation model shown in (3) by the random undersampling matrix Ξ, we can obtain the noiseless 1-D SAR undersampling observation model. It can be inferred that the random undersampled SAR echo $\mathbf{y}_{\mathrm{CS}} \in \mathbb{C}^I$ obtained by observing with the random observation matrix Φ_{CS} shown in (5) can also be expressed as $\mathbf{y}_{\mathrm{CS}} = \Xi\mathbf{y} = \Xi\Phi\mathbf{x} = \Phi_{\mathrm{CS}}\mathbf{x}$. The process of observing the SAR echo using the random observation matrix Φ_{CS} constructed based on (5) can be understood as follows: 1) on one hand, it can be understood as retaining some randomly selected sample points from the echo that has undergone complete Nyquist sampling, while discarding the rest of them; and 2) on the other hand, it can be understood as stopping the sampling of some randomly selected Nyquist sampling points based on the original Nyquist sampling-based echo observation, thereby reducing the average sampling rate of the entire sampling process. If we consider the additive noise that exists in the echo observation and recording process, the SAR undersampling echo observation model based on CS can be expressed as shown in the following equation:

$$\mathbf{y}_{\mathrm{CS}} = \Phi_{\mathrm{CS}}\mathbf{x} + \mathbf{n} = \Phi_{\mathrm{CS}}\Psi\mathbf{s} + \mathbf{n} \quad (6)$$

where $\mathbf{n} \in \mathbb{C}^I$ represents the additive noise, Ψ represents the sparse transformation basis matrix of the image $\mathbf{x}$, and $\mathbf{s}$ represents the corresponding sparse coefficients of the image.

III. Imaging Model and Its Iteration Solution

In this section, we will propose an SAR image-representation-constrained reconstruction model and provide its iterative solution based on PGD.

According to the CS theory, assuming that $\mathbf{n}$ follows a spherical Gaussian distribution, the reconstruction of the SAR image $\mathbf{x}$ can be achieved by solving the optimization problem shown in the following equation:

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}} \frac{1}{2}\|\Phi_{\mathrm{CS}}\mathbf{x} - \mathbf{y}_{\mathrm{CS}}\|_2^2 + \lambda\left\|\Psi^+\mathbf{x}\right\|_p^p \quad (7)$$

$$\begin{aligned}
\Phi &= [\phi(1,1), \phi(2,1), \ldots, \phi(P,1), \phi(1,2), \ldots, \phi(P,2), \ldots, \phi(P,Q)]^{\mathrm{T}} \\
\phi(p,q) &= \left[h_{1,1}(p,q), \ldots, h_{M,1}(p,q), \ldots, h_{1,N}(p,q), \ldots, h_{M,N}(p,q)\right]^{\mathrm{T}} \\
h_{m,n}(p,q) &= w_r\left(\tau_p - \tau_q(m,n)\right) w_a(t_q) \exp\left\{j\pi\gamma\left[\tau_p - \tau_q(m,n)\right]^2 - j2\pi f_c\tau_q(m,n)\right\}
\end{aligned} \quad (4)$$

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where $\Psi^{+}\Psi = \mathbf{I}_T$, $\mathbf{I}_T$ represents the T-dimensional identity matrix, and $\|\cdot\|_p$ denotes the p-norm with $0 \leqslant p \leqslant 1$. Equation (7) is the traditional reconstruction model for undersampled SAR imaging. For (7), one of the most common cases is when $p = 1$, which means using the l_1-norm to constrain the sparsity of the image. In this case, the model can be solved using the ISTA, which can reduce computational complexity [41]. Such traditional CS reconstruction models assume the sparsity of the scene or its transform domain, and therefore, they cannot perform well on nonsparse imaging scenes. The following content will focus on improving the imaging model to overcome this drawback.

In model construction, we abandoned the use of the regularization term based on sparse prior in (7) to constrain the model. Instead, to fully use the prior information of image representation, we further improved the l_p-norm term in the regularization term and proposed an image-representation-based model for undersampled SAR image reconstruction, as shown in the following equation:

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}} \frac{1}{2}\|\Phi_{\mathrm{CS}}\mathbf{x} - \mathbf{y}_{\mathrm{CS}}\|_2^2 + \sum_{t=1}^{T} \beta_t \, p_t(\mathbf{B}_t\mathbf{x}) \tag{8}$$

where $\beta_t = \rho\alpha_t$, $\mathbf{B}_t$ and p_t represent the feature transformation matrix and regularization function, respectively. It can be seen that the constraint on the imaging objective function here is no longer based on the sparsity prior of the scene, but instead uses a more generalized image representation for constraints. Therefore, the imaging model can be applied to more general imaging scenes, including both sparse and nonsparse imaging scenes, instead of just sparse scenes as before.

It is worth noting that in the image-representation-constrained reconstruction model shown in (8), both the feature transformation matrix $\mathbf{B}_t$ and the image-representation-constrained regularization function $p_t(\cdot)$ are undetermined. Therefore, for the objective function in (8), the l_2-norm term is differentiable, while the regularization term's differentiability is unknown. For optimization problems involving objective functions composed of both differentiable and nondifferentiable parts, an effective iterative optimization method called PGD method is proposed in [42], which can be viewed as an extension of ISTA. Based on the generalized model for undersampled SAR image reconstruction shown in (8), the corresponding PGD reconstruction method is derived as follows. If we consider the l_2-norm error term in (8) as a function of $\mathbf{x}$, denoted as $h(\mathbf{x}) = (1/2)\|\Phi_{\mathrm{CS}}\mathbf{x} - \mathbf{y}_{\mathrm{CS}}\|_2^2$, then it can be approximated by the following second-order Taylor expansion form in the neighborhood of $\mathbf{x}_k$, as shown in the following equation:

$$\begin{aligned}\hat{h}(\mathbf{x}) &= h(\mathbf{x}_k) + \mathrm{Re}\left([\nabla h(\mathbf{x}_k)]^{\mathrm{H}}(\mathbf{x} - \mathbf{x}_k)\right) + \frac{L}{2}\|\mathbf{x} - \mathbf{x}_k\|_2^2 \\ &= \frac{L}{2}\left\|\mathbf{x} - \left[\mathbf{x}_k - \frac{1}{L}\nabla h(\mathbf{x}_k)\right]\right\|_2^2 + C(\mathbf{x}_k)\end{aligned} \tag{9}$$

where $\nabla h(\mathbf{x}_k)$ represents the gradient of $h(\mathbf{x})$ at $\mathbf{x}_k$, and it can be expressed as $\nabla h(\mathbf{x}_k) = \Phi_{\mathrm{CS}}^{\mathrm{H}}(\Phi_{\mathrm{CS}}\mathbf{x}_k - \mathbf{y}_{\mathrm{CS}})$. Here, $L \in \mathbb{R}^{+}$ is a positive real constant, and $C(\mathbf{x}_k) = h(\mathbf{x}_k) - (1/2L)\|\nabla h(\mathbf{x}_k)\|_2^2$ is a constant term that depends on $\mathbf{x}_k$ but is independent of $\mathbf{x}$. Note that (9) is a quadratic function, and its minimum point $\mathbf{x}_{k+1}$ can be represented by the following equation:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \mu\nabla h(\mathbf{x}_k). \tag{10}$$

The above equation is exactly the iterative expression of gradient descent with a step size of $\mu = 1/L$. Equation (10) indicates that each iteration of gradient descent is equivalent to finding the minimum value of the second-order Taylor expansion of the objective function near the current iteration variable. Therefore, a similar approach can be considered to solve (8).

Using the approximate form of (9) from the previous expression, (8) can be represented as follows, in

$$\mathbf{x}_{k+1} = \arg\min_{\mathbf{x}} \frac{L}{2}\|\mathbf{x} - [\mathbf{x}_k - \mu\nabla h(\mathbf{x}_k)]\|_2^2 + \sum_{t=1}^{T} \beta_t \, p_t(\mathbf{B}_t\mathbf{x}) \tag{11}$$

where the constant term $C(\mathbf{x}_k)$ in (9) that is independent of $\mathbf{x}$ is ignored as it does not affect the solution to the optimization problem. Note that the gradient descent operation portion in the first term of (11) depends on $\mathbf{x}_k$ but is independent of $\mathbf{x}$. Therefore, (11) can be represented in the form of, as shown below

$$\mathbf{x}_{k+1} = \arg\min_{\mathbf{x}} \frac{1}{2}\|\mathbf{x} - \mathbf{z}_k\|_2^2 + \rho\sum_{t=1}^{T} \beta_t \, p_t(\mathbf{B}_t\mathbf{x}) \tag{12}$$

where $\mathbf{z}_k = \mathbf{x}_k - \mu\nabla h(\mathbf{x}_k)$. It can be observed that the right-hand side of the equation can be viewed as a transformation of $\mathbf{z}_k$ that depends on the parameter $\rho = 1/L$ and the regularization function. This transformation is known as the proximal operator and can be denoted as shown in the following equation:

$$\mathrm{prox}_{\rho g(\cdot)}(\mathbf{z}) = \arg\min_{\mathbf{x}} \frac{1}{2}\|\mathbf{x} - \mathbf{z}\|_2^2 + \rho\sum_{t=1}^{T} \beta_t \, p_t(\mathbf{B}_t\mathbf{x}). \tag{13}$$

Based on the above derivations, the PGD iterative solution to (8) is shown in the following equation:

$$\mathbf{z}_{k+1} = \mathbf{x}_k - \mu\Phi_{\mathrm{CS}}^{\mathrm{H}}(\Phi_{\mathrm{CS}}\mathbf{x}_k - \mathbf{y}_{\mathrm{CS}}) \tag{14}$$

$$\mathbf{x}_{k+1} = \mathrm{prox}_{\rho g(\cdot)}(\mathbf{z}_{k+1}) \tag{15}$$

where μ and ρ are the iteration parameters. The specific expression of the proximal operator can be calculated based on the specific form of the regularization function.

The general model for undersampled SAR reconstruction based on image representation is derived, and its iterative solution is given using the PGD algorithm. However, imaging using iterative solutions still cannot avoid long iteration times and manual parameter tuning, which needs to be further optimized in subsequent studies.

IV. NSR-NET for Nonsparse Scene

The deep unrolling network is a type of deep neural network designed based on an iterative algorithm, where the structure is equivalent to "unrolling" the iterative operation data flow that is repeatedly fed back into a gradually forwardfeeding neural

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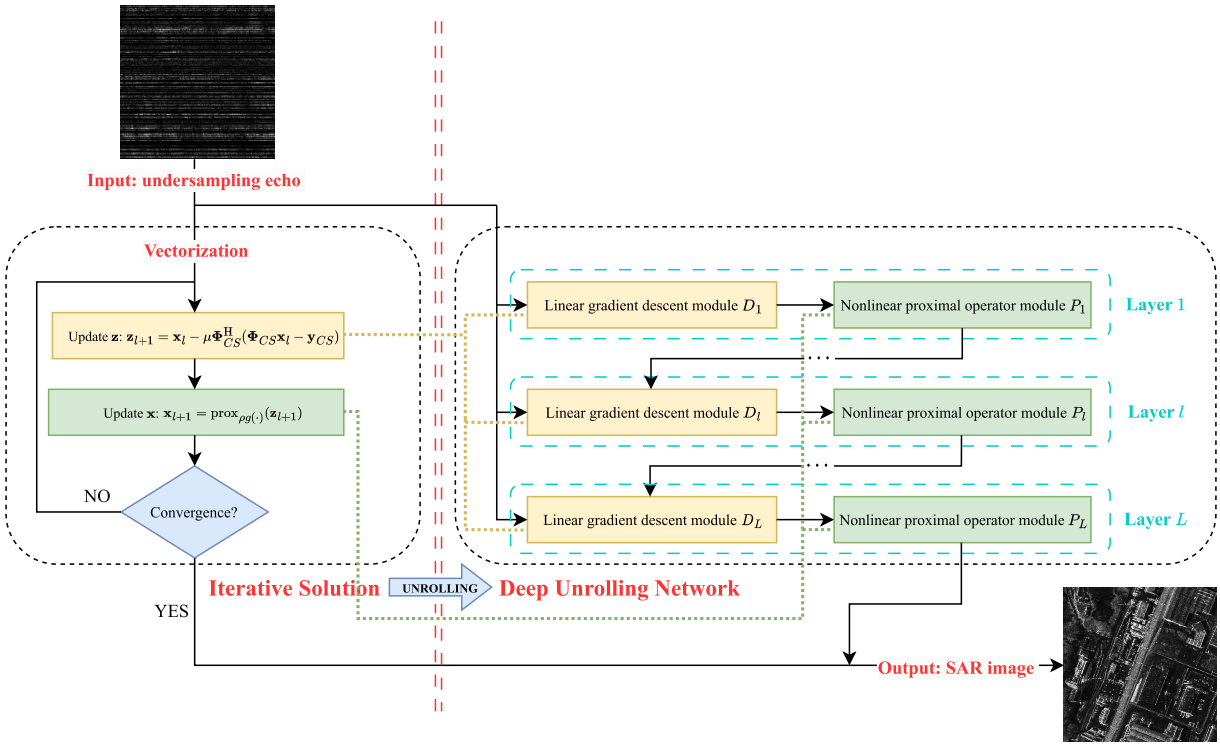

Fig. 2. Overall structure of the NSR-NET.

network data flow. Considering that (14) and (15) contain both linear and nonlinear operations, this matches the basic structural characteristics of deep neural networks. Therefore, in this section, we construct a nonsparse scene reconstruction network NSR-NET by setting each iteration step in (14) and (15) as a layer with learnable parameters in the network to learn the image representation from the dataset. Each layer of the network corresponds to the set of all the steps in one iteration. A finite number of network layers are then cascaded to form the entire deep unrolling network architecture, which can effectively address the difficulties encountered in iterative solutions.

The remaining content of this section will respectively introduce the overall structure of the network, the implementation details of each module and submodule, and specific implementation of the NSR-NET.

A. Overall Structure of the NSR-NET

Based on the characteristics of the iteration of PGD, the gradient descent operation step shown in (14) corresponds to the gradient descent module $D_l(l = 1, 2, \ldots, L)$, and the proximal operator operation step shown in (15) corresponds to the proximal operator module $P_l(l = 1, 2, \ldots, L)$ in the network after unrolling. The modules corresponding to these two operations can be repeated and cascaded L times to form the deep unrolling network, as shown in Fig. 2. One of the important differences between the deep unrolling network and the iterative method is that the parameters in the iterative algorithm can be set as learnable network parameters θ. For example, μ, ρ, feature extraction matrix $\mathbf{B}_t$, regularization function p_t, and so on in (14) and (15) are set as learnable parameters. This allows the parameters to be learned from the data through the network training process, obtaining the optimal parameters for each layer, and avoiding the tedious process of manual parameter tuning in the traditional iterative algorithms.

From Fig. 2, it can also be seen that the data processing flow of the above unrolling network can be regarded as a fixed number of iterations, where each iteration adjusts the iteration parameters adaptively according to the data features, making the iterative algorithm data processing flow converge rapidly. Another strong advantage of the deep unrolling network over the iterative method is that it can further introduce more network parameters beyond the iteration parameters by adjusting the original data processing method or designing new learnable modules based on the original data processing flow. This also enables the deep unrolling network to have far greater data adaptability than the iterative method. Compared with the iterative method, the deep unrolling network can achieve convergence with fewer equivalent iterations while obtaining better performance.

The NSR-NET is obtained by solving a model constrained by SAR image representation and unrolling it into a deep neural network. It not only possesses the advantages mentioned above regarding network unrolling but also, compared with the traditional sparse reconstruction networks, has the ability to learn more general SAR image representations beyond sparse characteristics. Therefore, the proposed network is applicable to both sparse SAR scene reconstruction and nonsparse SAR scene reconstruction, while sparse reconstruction networks are only suitable for reconstructing sparse scenes.

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B. Implementation Details of Each Module and Submodule

This section will introduce the implementation details of each module and submodule of the SAR imaging unrolling network based on image representation.

The linear operation performed in the gradient descent layer D_l is shown in the following equation:

$$\mathbf{z}_l = \mathbf{x}_{l-1} - \mu_l \Phi_{\mathrm{CS}}^{\mathrm{H}}(\Phi_{\mathrm{CS}}\mathbf{x}_{l-1} - \mathbf{y}_{\mathrm{CS}}). \tag{16}$$

The above equation is consistent with (14) in the iterative steps from Section III, which corresponds to the gradient descent operation of the l_2-norm term in (8), where μ_l is the step size of the update operation and is a learnable network parameter of this layer. When $l = 1$, the input update initial value $\mathbf{x}_0$ of this layer is set to all zeros.

The following content will introduce the implementation details of the proximal operator module P_l and its submodules. According to the definition of the proximal operator given in (13), and combined with the specific form of the regularization function in the image representation constrained reconstruction model, the nonlinear operation steps of the proximal operator shown in (15) can be specifically expressed as the form shown in the following equation:

$$\mathbf{x}_k = \arg\min_{\mathbf{x}} \frac{1}{2}\|\mathbf{x} - \mathbf{z}_k\|_2^2 + \sum_{t=1}^{T} \beta_t\, p_t(\mathbf{B}_t\mathbf{x}). \tag{17}$$

Since the form of the regularization function $p_t(\cdot)$ is unknown, it is difficult to directly obtain the expression of the solution to (17).

However, if we assume that $p_t(\cdot)$ is differentiable, we can consider using a finite number of gradient descent iterations to obtain an approximate solution [27]. The specific update steps are shown in the following equation:

$$\begin{aligned}\mathbf{x}_{k,r} &= \mathbf{x}_{k,r-1} - \eta\left[\mathbf{x}_{k,r-1} - \mathbf{z}_k + \sum_{t=1}^{T}\beta_t\mathbf{B}_t^{\mathrm{H}}\nabla p_t(\mathbf{B}_t\mathbf{x}_{k,r-1})\right] \\ &= (1-\eta)\mathbf{x}_{k,r-1} + \eta\mathbf{z}_k - \eta\nabla g(\mathbf{x}_{k,r-1})\end{aligned} \tag{18}$$

where $r = 1, 2, \ldots, N_r$, where N_r is the number of update iterations in the gradient descent method. $\mathbf{x}_{k,0} = \mathbf{z}_k$ represents the initial value of the update, and $\mathbf{x}_{k,N_r} = \hat{\mathbf{x}}_k$ represents the approximate solution of $\mathbf{x}_k$. η is the step size of the update operation, and $\nabla p_t(\cdot)$ represents the gradient mapping of $p_t(\cdot)$ with respect to its independent variable, which outputs the gradient value at the input complex vector. $\nabla g(\mathbf{x}) = \sum_{t=1}^{T}\beta_t\mathbf{B}_t^{\mathrm{H}}\nabla p_t(\mathbf{B}_t\mathbf{x})$ represents the gradient of the regularization term.

If we further assume that $p_t(\cdot)$ is a summation of the same operation performed independently on each component of the input, which has a similar property to the p-norm raised to the power of p, then $\nabla p_t(\cdot)$ can be expressed as a pointwise univariate function operation. If the undetermined univariate function is determined, then $\nabla p_t(\cdot)$ can be determined, and $p_t(\cdot)$ can be further determined. Furthermore, considering the commonly used ReLU function and other pointwise nonlinear layers in deep neural networks, which can be expressed as piecewise linear functions, we can use the following undetermined piecewise linear function shown in (19) to approximate the undetermined univariate function

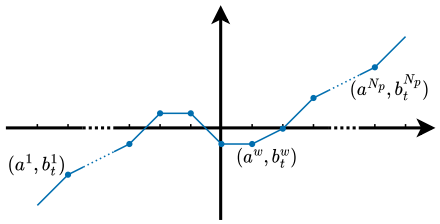

Fig. 3. Schematic of the piecewise linear function shown in (19).

$$s_t(x) = \begin{cases} x + b_t^1 - a^1, & x < a^1 \\ x + b_t^{N_p} - a^{N_p}, & x \geqslant a^{N_p} \\ b_t^w + \dfrac{b_t^{w+1} - b_t^w}{a^2 - a^1}(x - a^w), & a^w \leqslant x < a^{w+1} \end{cases} \tag{19}$$

where a^w and b_t^w for $w = 1, 2, \ldots, N_p$ represent the control points of the piecewise linear function in the horizontal and vertical directions, respectively, where N_p is the number of control points. Here, the horizontal coordinates are set to fixed values with equidistant distribution, and the vertical coordinates are undetermined. The schematic of this piecewise linear function is shown in Fig. 3.

In addition, according to the definition of 2-D convolution operation and its practical application, it can be seen that the transpose convolution operation $\mathbf{B}_t^{\mathrm{H}}$ in (18) can also be implemented by another convolution operation $\mathbf{B}_t'$. Therefore, the proximal operator operation shown in (18) can also be implemented using the construction idea of unrolling network. Here, $\nabla g(\mathbf{x}_{k,r-1})$ can be decomposed into two 2-D convolution operations interleaved with a pointwise piecewise linear function operation, and then completed by a weighted sum operation to perform one update operation. By combining the corresponding network structures in the order of operations and repeating the concatenation for N_r times, it can be used as a specific implementation of the proximal operator nonlinear layer.

In summary, the N_r step gradient descent update operation is implemented to approximate the proximal operator operation in (17). Each step of the gradient descent update operation in the proximal operator layer P_l can be expressed as follows:

$$\begin{aligned}\mathbf{x}_{l,r} &= (1 - \eta_{l,r})\mathbf{x}_{l,r-1} + \eta_{l,r}\mathbf{z}_l - \eta_{l,r}\nabla g(\mathbf{x}_{l,r-1}) \\ \nabla g(\mathbf{x}_{l,r-1}) &= \sum_{t=1}^{T}\mathbf{B}_{t,l,r}^2 S_{t,l,r}(\mathbf{B}_{t,l,r}^1\mathbf{x}_{l,r-1})\end{aligned} \tag{20}$$

where the 2-D convolution operations $\mathbf{B}_{t,l,r}^1$ and $\mathbf{B}_{t,l,r}^2$ correspond to $\mathbf{B}_t$ and $\mathbf{B}_t^{\mathrm{H}}$ in (18).

C. Specific Implementation of the NSR-NET

All the derivations above are based on a 1-D imaging model, in which the data are processed in the form of vectors. However, in reality, the data in the network are processed in 2-D form to improve computational efficiency. The following content will introduce the specific implementation of the NSR-NET.

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The first part is about the approximation of the observation matrix. In the gradient descent layer operation shown in (16), an undersampling SAR observation matrix Φ_{CS} is used. However, directly using Φ_{CS} leads to low computational efficiency, which significantly reduces the speed of the deep unrolling network that needs to perform this operation on a large amount of data multiple times. To avoid using the above observation matrix directly and improve computational efficiency, an approximate observation operator in the 2-D form [11] can be used in the network operation.

Specifically, to improve computational efficiency, one can use the linear mapping operator G and its inverse operator M based on matched filtering (MF) as approximations of the observation matrices Φ^{H} and Φ, respectively, under the condition that the operation performed by the imaging algorithm is reversible. The MF-based method used in this article is the chirp scaling algorithm (CSA), and the specific form of the inverse approximate observation operator M is in (21), as shown at the bottom of the page, which can achieve the mapping from echoes to echo SAR image based on CSA. where $\mathcal{F}_a\{\cdot\}$ and $\mathcal{F}_r\{\cdot\}$ represent the Fourier transform operators in azimuth and range, respectively, and $\mathcal{F}_a^{-1}\{\cdot\}$ and $\mathcal{F}_r^{-1}\{\cdot\}$ are the corresponding inverse operations. $\Phi_i(i=1,2,3)$ are the phase terms used for range migration correction, residual phase compensation, etc. G represents the corresponding approximate observation operator and satisfies $\mathbf{S} \approx G\{\mathbf{X}\}$.

Then the 2-D implementation of (16) in the network is shown in the following equation:

$$\mathbf{Z}_l = \mathbf{X}_{l-1} - \mu_l M\{G_{\mathrm{CS}}(\mathbf{X}_{l-1}) - \mathbf{S}_{\mathrm{CS}}\} \tag{22}$$

where G_{CS} represents the undersampled inverse chirp scaling operator, and S_{CS} represents the undersampled echo matrix.

Second, we will introduce the implementation details of the NSR-NET, which are shown in Fig. 4, and $\mathbf{B}_{t,l,r}^1$ and $\mathbf{B}_{t,l,r}^2$ are implemented by multichannel 2-D convolution layers $C_{l,r}^1$ and $C_{l,r}^2$ in the network architecture, respectively. The pointwise piecewise linear function operation $S_{t,l,r}(\cdot)$ is used to implement $\nabla p_t(\cdot)$ in (18). This operation is implemented by a multichannel nonlinear layer $H_{l,r}$ in the network architecture. After computing $\nabla g(\mathbf{x}_{l,r-1})$ using the above network structure, the final output $\mathbf{x}_{l,r}$ of the update operation is calculated by a weighted sum layer $A_{l,r}$, where $\eta_{l,r}$ is a learnable parameter of this layer.

It should be noted that $\nabla g(\mathbf{x}_{l,r})$ in (20) is in the form of multiple summations, which can be naturally implemented by a multichannel operation. For example, in the multichannel 2-D convolution layer $C_{l,r}^1$, T single-channel 2-D convolution kernels are used to perform convolution operations corresponding to $\mathbf{B}_{t,l,r}^1$ on the single-channel input $\mathbf{x}_{l,r-1}$, resulting in T output channels. The multichannel nonlinear layer $H_{l,r}$ performs pointwise piecewise linear function operations on the data of different channels separately. The multichannel 2-D convolution layer $C_{l,r}^2$ uses one T-channel convolution kernel to perform convolution operations corresponding to $\mathbf{B}_{t,l,r}^2$ on each channel of the input T-channel data separately and then combines the results of each channel by summation to obtain a single-channel output. Considering that some network structures used to implement $\nabla g(\mathbf{x}_{l,r})$, such as $H_{l,r}$ and $C_{l,r}^2$, can achieve the function of weighted summation of different channels through their learnable parameters, the weighting coefficients β_t in (18) are omitted in the network operation shown in (20).

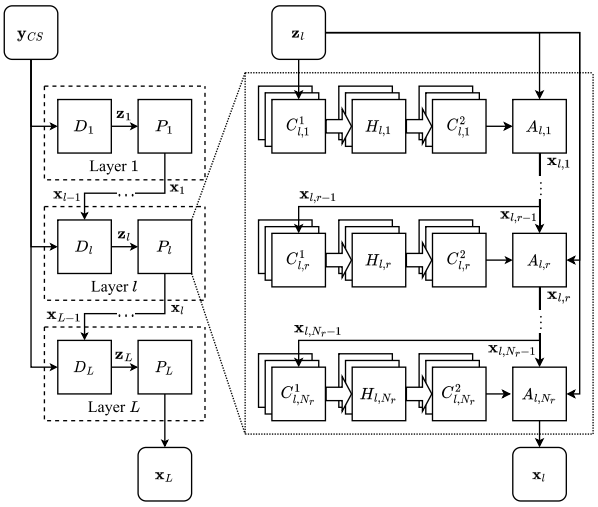

Fig. 4. Implementation details of the NSR-NET.

When performing network imaging, the processing of complex-valued data is involved. Next, we will use the convolution layer as an example to introduce the method of processing complex-valued data. In the proximal operator layer operation shown in (20), since the input of the 2-D convolution operation $\mathbf{B}_{t,l,r}^1, \mathbf{B}_{t,l,r}^2$ can be complex, it is also necessary to consider the case where the convolution kernel is complex. According to [43], let $\mathbf{B}_{\mathbf{W}}$ represent the 2-D convolution matrix of the complex convolution kernel $\mathbf{W}$, let $\mathbf{X}$ represent the complex image undergoing the 2-D convolution operation, and let $\mathbf{x} = \mathrm{vec}(\mathbf{X})$. Then, according to the rules of complex arithmetic, the following complex convolution operation rule can be obtained as shown in

$$\begin{aligned}\mathbf{B}_{\mathbf{W}}\mathbf{x} = &\left[\mathbf{B}_{\mathbf{W}}^{\mathrm{R}}\,\mathrm{Re}(\mathbf{x}) - \mathbf{B}_{\mathbf{W}}^{\mathrm{I}}\,\mathrm{Im}(\mathbf{x})\right] \\ &+ j\left[\mathbf{B}_{\mathbf{W}}^{\mathrm{R}}\,\mathrm{Im}(\mathbf{x}) + \mathbf{B}_{\mathbf{W}}^{\mathrm{I}}\,\mathrm{Re}(\mathbf{x})\right]\end{aligned} \tag{23}$$

where $\mathbf{B}_{\mathbf{W}}^{\mathrm{R}} = \mathrm{Re}(\mathbf{B}_{\mathbf{W}}), \mathbf{B}_{\mathbf{W}}^{\mathrm{I}} = \mathrm{Im}(\mathbf{B}_{\mathbf{W}})$. It is also known that $\mathbf{B}_{\mathbf{W}}^{\mathrm{R}}, \mathbf{B}_{\mathbf{W}}^{\mathrm{I}}$ are real convolution matrices with real and imaginary parts of $\mathbf{W}$ as kernels, respectively, denoted by $\mathbf{W}^{\mathrm{R}} = \mathrm{Re}(\mathbf{W})$ and $\mathbf{W}^{\mathrm{I}} = \mathrm{Im}(\mathbf{W})$. Then, from (23), it can be seen that a complex convolution layer $C^{\mathbb{C}}$ with a kernel of $\mathbf{W}$ can be implemented by two real convolution layers $C^{\mathrm{R}}, C^{\mathrm{I}}$ with

$$\mathbf{X} = M\{\mathbf{S}\} = \mathcal{F}_a^{-1}\left\{\mathcal{F}_r^{-1}\left\{\mathcal{F}_r\left\{\begin{matrix}\mathcal{F}_a\{\mathbf{S}\} \\ \cdot \exp\left[j\Phi_1(f_a, f_r)\right]\end{matrix}\right\} \cdot \exp\left[j\Phi_2(f_a, f_r)\right]\right\} \cdot \exp\left[j\Phi_3(f_a, f_r)\right]\right\} \tag{21}$$

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kernels of $\mathbf{W}^{\mathrm{R}}, \mathbf{W}^{\mathrm{I}}$, respectively. When $C^{\mathbb{C}}$ performs complex convolution operation, the calculation can be done following the rule shown in (23).

In the specific implementation of the convolutional layer, after completing the convolution operation, a per-point scalar bias is added to each channel output separately. The 2-D operation can be represented as $\mathbf{S} = \mathbf{X} * \mathbf{W} + \mathbf{b}_{\mathrm{mat}}$, where the bias matrix $\mathbf{b}_{\mathrm{mat}}$ has the same size as the convolution output, but all the elements are learnable scalar bias values b. Adding bias can adjust the importance of different channels to some extent, similar to the effect of the weighting coefficient β_t in (18). In the complex convolution layer $C^{\mathbb{C}}$ implemented in the above way, if both the real convolution layers $C^{\mathrm{R}}, C^{\mathrm{I}}$ have added bias, according to the operation rule shown in (23), the equivalent bias of $C^{\mathbb{C}}$ is shown in the following equation:

$$\mathbf{b}_{\mathrm{vec}}^{\mathbb{C}} = (\mathbf{b}_{\mathrm{vec}}^{\mathrm{R}} - \mathbf{b}_{\mathrm{vec}}^{\mathrm{I}}) + j\,(\mathbf{b}_{\mathrm{vec}}^{\mathrm{R}} + \mathbf{b}_{\mathrm{vec}}^{\mathrm{I}}) \tag{24}$$

where $\mathbf{b}_{\mathrm{vec}}^{\mathbb{C}}, \mathbf{b}_{\mathrm{vec}}^{\mathrm{R}}, \mathbf{b}_{\mathrm{vec}}^{\mathrm{I}}$, respectively, represent the bias vectors in the complex convolution layer $C^{\mathbb{C}}$ and the real convolution layers $C^{\mathrm{R}}, C^{\mathrm{I}}$, where each element corresponds to the bias value of each channel.

V. Experimental Results and Discussions

This section mainly introduces the related simulation experiments of the NSR-NET based on image representation, presents the experimental data acquisition method and the corresponding simulation parameters, provides the test results of the network under different conditions, and analyzes the experimental effects.

A. Experimental Dataset and Corresponding Simulation Parameters

For the training and testing of the proposed network, a large number of image–echo data pairs are required as training samples. The echo data can be generated from SAR images in the dataset. After generating the complete echo data corresponding to SAR images in the dataset, undersampling is applied to the complete echo data. Given that random undersampling in the range direction may cause significant difficulties in the acquisition of A/D signals and hardware design in practical applications, the undersampling method we used preserves all the rangewise echo signals and randomly retains a certain proportion of the azimuthwise echo signals. The value of the proportion is referred to as the undersampling rate. The SAR samples in the dataset and undersampled echoes are combined to form image–echo data pairs, which can be used for training and testing the deep unrolling network for nonsparse SAR imaging.

The ground-truth scenes used for the NSR-NET and other comparative methods are the same and sourced from TerraSAR-X, an Earth observation satellite that provides a wealth of open-source high-resolution SAR images. The training dataset, in particular, is cropped from a high-resolution image of the Tucson area, which represents an urban environment with diverse scene distributions, exhibiting prominent characteristics of nonsparse scenes. Each sample image has a size of 512 pixels in both width and height ($M = N = 512$),

Fig. 5. Example of simulated samples in the dataset.

TABLE I
SAR System Parameters Used in Simulation Experiments

Parameter	Value
Carrier frequency f_c	10 GHz
Pulse repetition frequency f_{PRF}	1000 Hz
Transmitted signal bandwidth B	300 MHz
Platform speed v	120 m/s
Pulse width T_p	1 μs
Closest slant range to scene center R_B	4 km
Fast time sampling frequency f_s	450 MHz

and a total of 300 samples were used for training ($S = 300$), 50 for validation, and 30 for testing. The two randomly selected samples are shown in Fig. 5. It can be seen that the scene contains densely arranged buildings and other objects, which occupy the majority of the scene. Therefore, it represents a typical nonsparse scene. Unless otherwise specified, the SAR system parameters used in the simulation experiments are shown in Table I.

For the setting of network parameters, the hyperparameters of the network architecture used in the experiment are set as follows: the convolution kernel size $M_{\mathbf{W}} = N_{\mathbf{W}} = 3$, the number of feature channels $T = 4$, the number of layers $L = 6$ in the network submodules, the number of iterations $N_r = 2$ for the gradient descent operation in each near-end operation layer P_l, and the values of the abscissae $\{a^w\}_{w=1}^{N_p}$ of the control points in the piecewise linear function are uniformly distributed in the interval $[-1, 1]$, with a total of $N_p = 21$ points. The initial values of the learnable parameters in the network are set as follows: the update step size parameters $\{\mu_l, \{\eta_{l,r}\}_{r=1}^{N_r}\}_{l=1}^{L}$ in each gradient descent layer D_l and weighted summation layer $A_{l,r}$ are all initialized to 0.1, and the abscissae $\{b_{t,l,r}^{\mathrm{R},w}, b_{t,l,r}^{\mathrm{I},w}\}_{w=1}^{N_p}$ of the control points for each channel in the nonlinear layer $H_{l,r}$ are initialized according to the ReLU function, while the other parameters such as the convolution kernels and biases in the 2-D convolution layers $C_{l,r}^1, C_{l,r}^2$ are initialized in the default manner.

Regarding network training, the loss function used for training network parameters is the l_2 loss function without regularization, i.e., $h(\hat{\mathbf{X}}, \mathbf{X}) = |\hat{\mathbf{X}} - \mathbf{X}|_2^2/\mathbf{S}$ and $g_{\mathrm{reg}}(\boldsymbol{\theta}) = 0$. The update optimization method for the parameters is batch-update RMSProp with a batch size of 10, a learning rate of 0.005, and a total of 4500 training iterations (i.e., 150 passes through the training set). For the training platform, a server container with

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a 12-core Intel Xeon Platinum 8255C CPU and one NVIDIA RTX 3090 GPU was used, with each training iteration taking about 1 h. The hardware configuration for the testing platform is an Intel[1] Core[2] i5-9300H CPU and one NVIDIA GeForce GTX 1050 GPU.

The performance metrics for measuring the SAR image reconstruction performance include the normalized mean square error (NMSE), peak signal-to-noise ratio (PSNR), and structural similarity index (SSIM). The definition of NMSE is given by the following equation:

$$\text{NMSE} = \frac{\left\|\mathbf{X} - \hat{\mathbf{X}}\right\|_F^2}{\left\|\mathbf{X}\right\|_F^2} \tag{25}$$

where $\mathbf{x}$ and $\hat{\mathbf{x}}$, respectively, represent the original SAR image and the reconstructed image output by the network. The definition of PSNR is given by the following equation:

$$\text{PSNR} = 20\log_{10}\frac{\max(|\mathbf{X}|)}{\sqrt{\frac{1}{MN}\left\|\mathbf{X} - \hat{\mathbf{X}}\right\|_F^2}} \tag{26}$$

where $\max(\cdot)$ denotes the maximum element operation. The definition of SSIM is given by the following equation:

$$\text{SSIM} = \frac{(2\mu_{\mathbf{X}}\mu_{\hat{\mathbf{X}}} + c_1)(2\sigma_{\mathbf{X}\hat{\mathbf{X}}} + c_2)}{(\mu_{\mathbf{X}}^2 + \mu_{\hat{\mathbf{X}}}^2 + c_1)(\sigma_{\mathbf{X}}^2 + \sigma_{\hat{\mathbf{X}}}^2 + c_2)} \tag{27}$$

where $\mu_{\mathbf{X}}$ and $\mu_{\hat{\mathbf{X}}}$ represent the mean values of the original and reconstructed images, respectively, while $\sigma_{\mathbf{X}}^2$ and $\sigma_{\hat{\mathbf{X}}}^2$ represent their variances, and $\sigma_{\mathbf{X}\hat{\mathbf{X}}}^2$ represents their covariance. The parameters c_1 and c_2 are small constants that satisfy $c_1 = (k_1 R)^2$ and $c_2 = (k_2 R)^2$, where k_1 and k_2 are constants typically set to 0.01 and 0.03, respectively, and R is the dynamic range of the image. All the metrics, except NMSE, are positively correlated with reconstruction performance.

B. Comparative Experiment of Different Methods Under Different Undersampling Rates

The experiments first compared the performance of different SAR image reconstruction methods under different undersampling rates. The compared methods compressed sensing reconstruction with dual-adaptive dictionary algorithm (CSRDA) [11], the RMIST-Net in [33], the LRSR-ADMM-Net in [39], and the CFPO-AutoImag-Net in [40]. All the training and testing data used in comparative methods were as described in Section V-A. The platform for all the comparative experiments is consistent with what is shown in Section V-A. The network parameters of the comparison methods are set according to the corresponding references. Fig. 6 shows two ground-truth images from the test data, corresponding to scenes 10 and 17. The reconstruction results are shown in Figs. 7 and 8, respectively. From Figs. 7 and 8, it can be observed that the proposed method demonstrates excellent reconstruction capability for nonsparse scenes at different undersampling rates. Even at a low undersampling rate of 30%, our method accurately reconstructs both sparse strong scattering targets and nonsparse scene details. In contrast,

Fig. 6. Ground-truth images used for SAR image reconstruction. The left and right correspond to scene 10 and scene 17, respectively.

CSRDA and RMIST-Net, which are based on sparse prior, can only reconstruct strong scattering targets in the scene at high undersampling rates, and they suffer from a degradation in visual quality when the amount of data is limited. LRSR-ADMM-Net achieves relatively accurate reconstruction at 70% and 50% undersampling rates but experiences a significant drop in performance at a 30% undersampling rate. CFPO-AutoImag-Net shows less accurate reconstruction for nonsparse backgrounds and also suffers from severe performance degradation with limited echo data. The average performance metrics of the 30 test samples using different methods are shown in Table II. From Table II, it can be seen that the proposed SAR image reconstruction method based on image representation outperforms all the compared methods, especially at a 30% undersampling rate, all the performance metrics of the comparative methods significantly decrease, while our proposed method still achieves a PSNR of 28.7286 and an SSIM of 0.9176, respectively.

In addition, Table III records the reconstruction time required for each method to reconstruct one image, averaged over five reconstructions at a 50% undersampling ratio. It can be observed that CSRDA typically requires tens or even hundreds of iterations to achieve optimal results, resulting in longer computation time compared with methods using unrolling networks. LRSR-ADMM-Net in [39] involves singular value decomposition (SVD) operations in the network, which incur significant computational costs. On the other hand, although our method consumes more time per computation due to the network structure and parameter size compared with RMIST-Net in [33] and CFPO-AutoImag-Net in [40], considering the tradeoff between computational time and imaging performance, the time difference is acceptable.

C. Transferability Validation Experiments of Different Radar Parameters and Imaging Scenes

While training deep neural networks can take a certain amount of time, if a well-trained network can be applied to imaging in different scenes and radar parameters with good imaging results, then we only need to train once and can continue to use it, which is very efficient. Therefore, the transferability validation experiments are designed to verify the transfer and generalization ability of the proposed SAR image reconstruction method based on image representation, which refers to the ability to adapt to some changes in the testing conditions without retraining the network. Specifically,

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[2]Trademarked.

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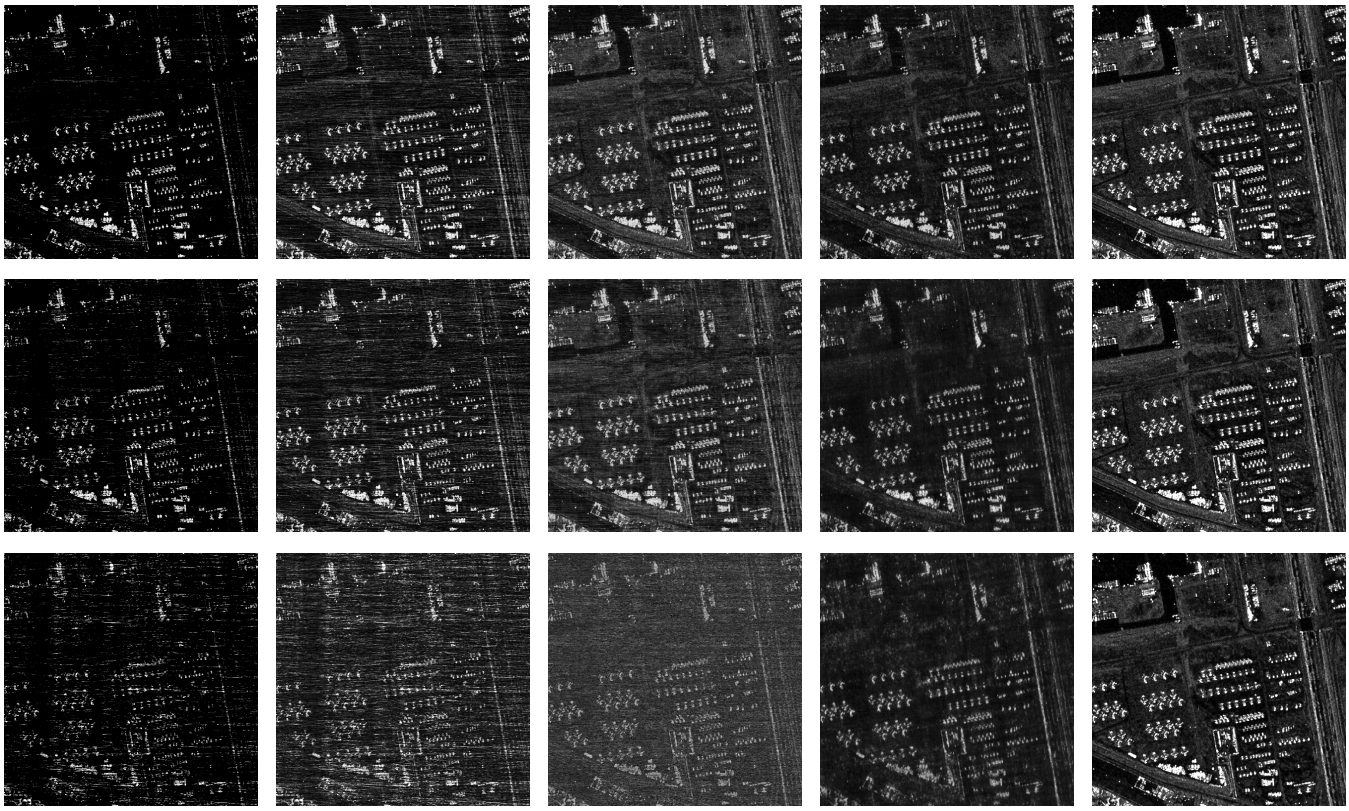

Fig. 7. Comparison of SAR scene ten reconstruction results under different sampling rates. From the first row to the third row, the results correspond to downsampling rates of 70%, 50%, and 30%. The columns from left to right represent the result using the CSRDA method, the RMIST-Net method, the LRSR-ADMM-Net method, the CFPO-AutoImag-Net method, and the proposed method, respectively.

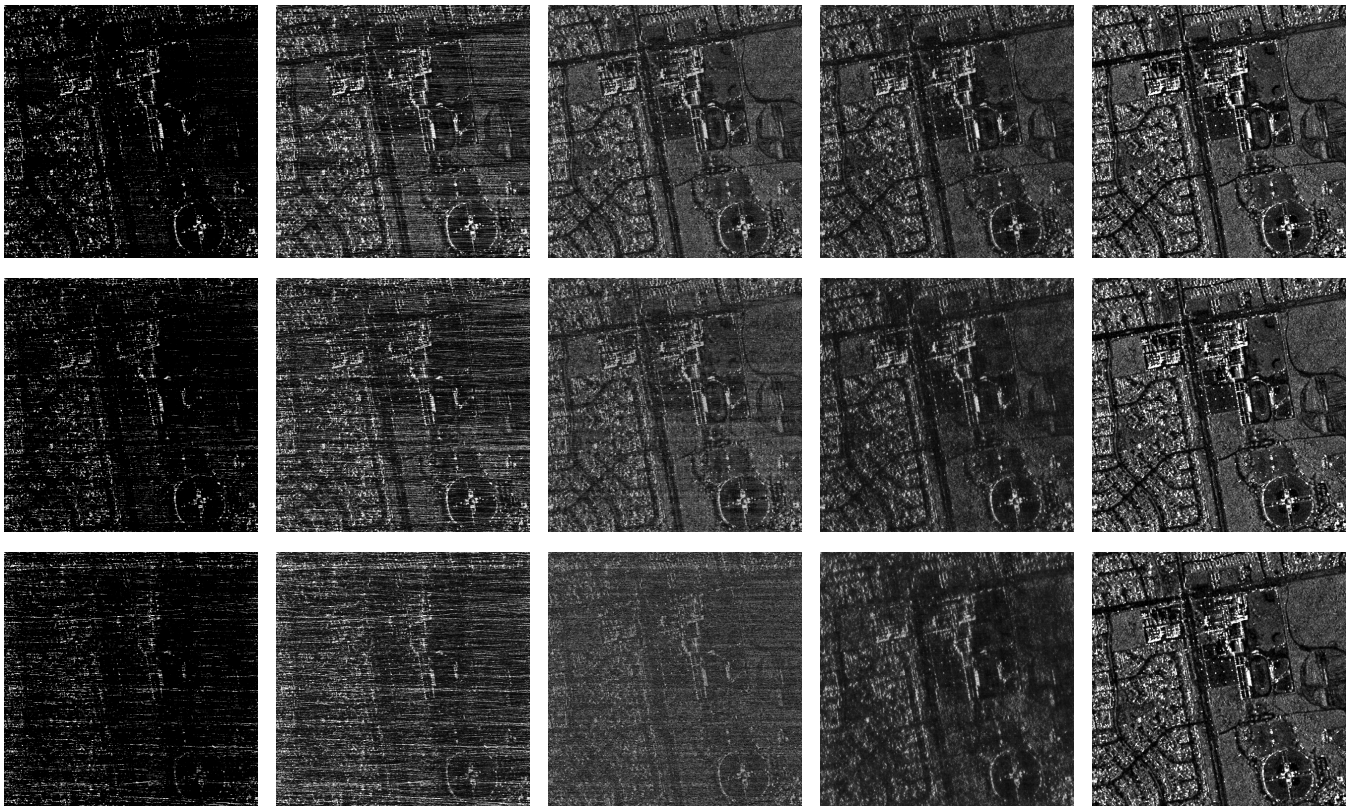

Fig. 8. Comparison of SAR scene 17 reconstruction results under different sampling rates. From the first row to the third row, the results correspond to downsampling rates of 70%, 50%, and 30%. The columns from left to right represent the result using the CSRDA method, the RMIST-Net method, the LRSR-ADMM-Net method, the CFPO-AutoImag-Net method, and the proposed method, respectively.

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TABLE II
COMPARISON OF SAR IMAGE RECONSTRUCTION PERFORMANCE METRICS AT DIFFERENT UNDERSAMPLING RATES

↓Rate	Performance Metric	CSDRA	RMIST-Net	Method in [39]	Method in [40]	NSR-NET
70%	NMSE	0.7095	0.1304	0.0279	0.0324	**0.0003**
	PSNR(dB)	11.8304	19.1826	25.9686	26.0870	**45.4130**
	SSIM	0.0793	0.6308	0.8894	0.7938	**0.9985**
50%	NMSE	0.7725	0.1964	0.0833	0.0710	**0.0024**
	PSNR(dB)	11.4368	17.4250	21.1955	23.1185	**36.5054**
	SSIM	0.0592	0.5059	0.7132	0.6293	**0.9853**
30%	NMSE	0.8409	0.2769	0.2048	0.1237	**0.0145**
	PSNR(dB)	11.0559	16.0236	17.2880	20.7423	**28.7286**
	SSIM	0.0449	0.3986	0.4922	0.4536	**0.9176**

TABLE III
COMPARISON TABLE OF RECONSTRUCTION TIME FOR DIFFERENT METHODS

Method	CSRDA	RMIST-Net	Method in [39]	Method in [40]	NSR-NET
Time(s)	7.1233	0.7325	5.3128	1.2028	**3.0849**

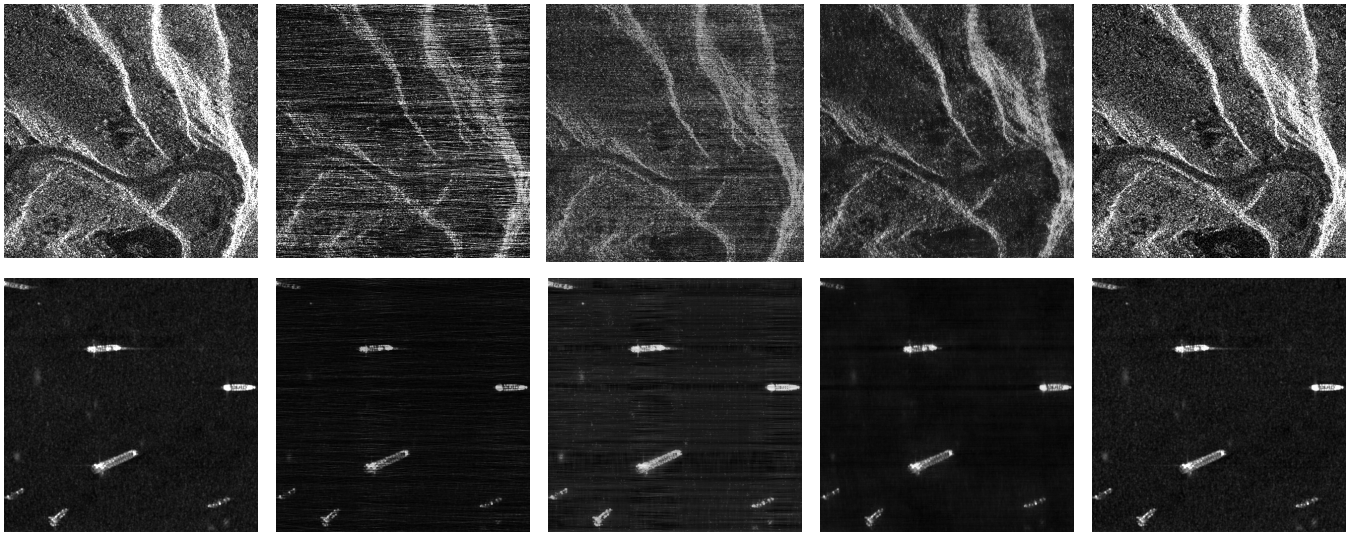

Fig. 9. Comparison of reconstruction results of the proposed method in the transferability validation experiment with changed scenes. The first row shows one of the test samples of the mountain terrain scene, and the second row shows one of the test samples of the sparse maritime scene. The columns from left to right represent the original image, the result using the RMIST-Net method, the result using the LRSR-ADMM-Net method, the result using the CFPO-AutoImag-Net method, and the result using the proposed method, respectively.

on one hand, the performance of the network trained on the original training dataset was tested on 30 test sample data from another SAR scene. On the other hand, the performance of the trained network was tested when changing the SAR system parameters. In the above tests, the undersampling ratio was fixed at 50%.

1) Transferability Validation Experiment With Changed Scenes: In the transferability validation experiments with changed scenes, we selected two typical scenes for testing to prove that the proposed method applies to both sparse and nonsparse scenes, namely, a mountain terrain scene with higher nonsparsity and a sparse maritime scene. In this experiment, the RMIST-Net, LRSR-ADMM-Net, and CFPO-AutoImag-Net are chosen as comparative methods, and the test results are presented in Table IV. Combining the samples from two different test sets shown in Fig. 9, it can be observed that the variation in test scene types has a slight impact on our proposed method, but it still demonstrates good reconstruction performance in both transfer scenes. Compared with the proposed method, the RMIST-Net is based on sparse prior modeling, which results in good reconstruction performance for the sparse maritime scene. However, it suffers from severe

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TABLE IV
PERFORMANCE COMPARISON TABLE OF TRANSFERABILITY VALIDATION EXPERIMENTS WITH CHANGED SCENES

Scene type	Performance Metric	RMIST-Net	Method in [39]	Method in [40]	NSR-NET
Mountain terrain	NMSE	0.2371	0.1301	0.1026	**0.0100**
	PSNR(dB)	14.0356	16.6966	19.0953	**27.7619**
	SSIM	0.4053	0.6838	0.5983	**0.9680**
Sparse maritime	NMSE	1.4696	0.1112	0.0814	**0.0018**
	PSNR(dB)	16.5076	25.6824	26.5939	**42.6464**
	SSIM	0.4995	0.8058	0.7206	**0.9934**

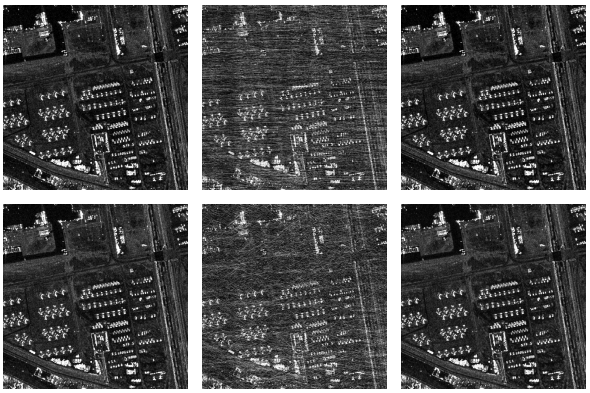

Fig. 10. Comparison of reconstruction results of scene 10 in the transferability validation experiment with changed parameters, where the zero-filling and direct MF imaging (ZF-MF) method is used to visually demonstrate the change in radar parameters. The first row shows the imaging results without changing the radar parameters, and the second row shows the imaging results after changing the radar parameters. and each column from left to right corresponds to the original image, the imaging result using the ZF-MF method, and the imaging result using the proposed network.

performance degradation in the nonsparse mountain terrain scene transfer experiment. LRSR-ADMM-Net, on the other hand, exhibits mediocre transfer capability for scenes with significant distribution changes due to its modeling limitation. CFPO-AutoImag-Net shows some scene transfer capability, but its performance is noticeably inferior compared with our proposed method.

2) Transferability Validation Experiment With Changed Radar Parameters: During training, all the training data have the same radar parameters. However, in reality, the platform may have different velocities, flight altitudes, bandwidths of the transmitted signals, and other parameters on different imaging tasks. Therefore, it is necessary to investigate the transferability of the network under changed parameters. In the transferability validation experiments with changed radar parameters, the detailed parameter changes are shown in Table V, including reducing the platform speed from 120 to 40 m/s and increasing the signal bandwidth from 300 to 800 MHz during signal transmission, among others, and the obtained reconstruction performance is shown in Table VI. The imaging result of scene 10 from the test data is shown in Fig. 10. It can be seen that changing the radar parameters of the test data only results in a slight decrease

TABLE V
SAR SYSTEM PARAMETERS USED IN THE TRANSFERABILITY VALIDATION EXPERIMENT

Parameter	Value
Carrier frequency f_c	10 GHz
Pulse repetition frequency f_{PRF}	2500 Hz
Transmit signal bandwidth B	800 MHz
Platform speed v	40 m/s
Pulse width T_p	1.5 μs
Minimum slant range to scene center R_B	1 km
Fast time sampling frequency f_s	1000 MHz

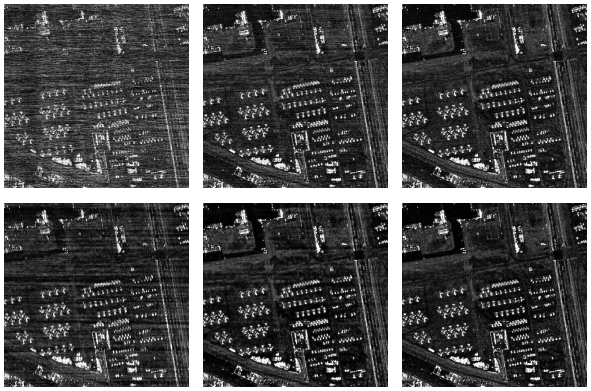

Fig. 11. Visualization comparison of image representation prior information used in the proposed network, where the top and bottom rows correspond to the input and output of the proximal operator modules, respectively, and each column from left to right corresponds to the proximal operator module for the first to the third layer of the network.

in imaging performance. Therefore, it can still be considered that the proposed network trained under one set of radar parameters can be directly applied in the imaging of other radar parameters. The explanation for this is that the proposed network takes the radar parameters as known information input into the network during the gradient descent module. Moreover, the proposed deep unrolling network fully learns the common SAR image representation, even if the radar parameters are adjusted, the common image representation of the SAR image still exists. Therefore, the network still performs well in the imaging task after changing the radar parameters.

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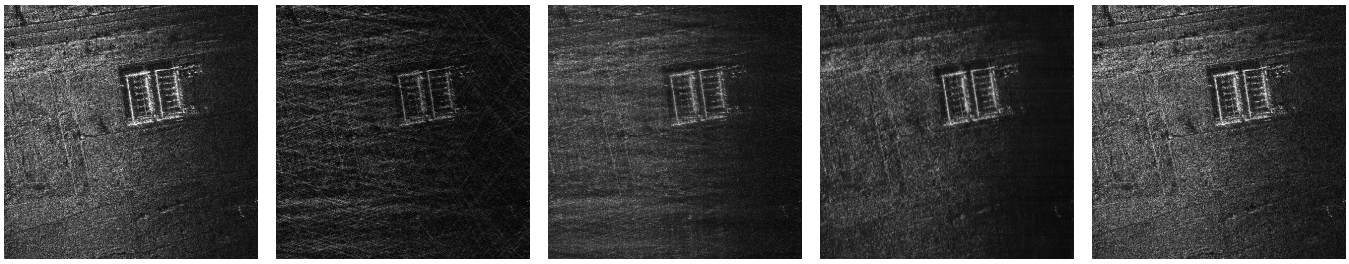

Fig. 12. Test results with measured data. The first image represents the imaging result using complete echo data, while the subsequent images depict the imaging results using RMIST-Net, LRSR-ADMM-Net, CFPO-AutoImag-Net, and the proposed method with undersampling rate of 50%.

TABLE VI
PERFORMANCE COMPARISON TABLE OF TRANSFERABILITY VALIDATION EXPERIMENTS WITH CHANGED RADAR PARAMETERS

Change or not	NMSE	PSNR(dB)	SSIM
NO	0.0024	36.5054	0.9853
YES	0.0043	34.0050	0.9729

D. Visualization Experiment for Image Representation Learning

Finally, to demonstrate the image representation prior used in the proposed network, we design a visualization experiment for image representation learning, Fig. 11 shows a comparison of the input and output of proximal operator modules in different layers under a 50% undersampling rate. By comparing the input and output of the proximal operator modules in Fig. 11, it can be seen that the main function of this module is to suppress the blurring caused by undersampled echo imaging, enhance the texture, edge, and other detailed features in the nonsparse part, and retain sparse bright points and speckle noise features in SAR images. And the improvement becomes increasingly evident as the number of network layers increases. This indicates that the proposed network learns and extracts various image representation priors in the image, including sparse features, and suppresses, enhances, or retains the above features according to the high-precision undersampled SAR imaging requirements.

E. Test With Measured Echo Data

To validate the effectiveness of the proposed method on measured echo data, we also conducted a test using real measured data. The measured data were obtained from an airborne SAR system with the following radar parameters: carrier frequency f_c = 9.6 GHz, pulse repetition frequency (PRF) of 3200 Hz, transmitted signal bandwidth of 400 MHz, platform velocity of 73 m/s, pulsewidth of 8×10^{-6} s, and minimum slant range from the platform to the scene center of 3754 m. After compensating for motion errors in the measured data, it was undersampled at a 50% undersampling rate, and the undersampled echo was used as input to the network. The comparative methods include the RMIST-Net, the LRSR-ADMM-Net, and the CFPO-AutoImag-Net, respectively. The imaging results are shown in Fig. 12, and it can be observed that the proposed method also produces good focusing effects on the measured echo data. In comparison, the imaging performance of the comparative methods is significantly lower than that of the proposed method when applied to the measured data. Compared with the result using complete echo data, the SSIM of the proposed method is calculated to be 0.9001, while the other three comparative methods have SSIM values of 0.5095, 0.6757, and 0.5450, respectively, which further confirms the superiority of the proposed method.

VI. CONCLUSION

This article proposes an SAR unrolling network imaging method that uses the general SAR image representation to solve the problem of undersampled SAR imaging in nonsparse scenes. First, based on the CS theory, a general image representation constraint model for undersampled SAR image reconstruction is established and the PGD algorithm was used to obtain its iterative solution. Then, the specific iterative solution steps are unrolled into a deep network named NSR-NET according to the relevant theoretical research on DL, especially deep unrolling networks. In the unrolling network, the basic idea of using 2-D convolution to extract SAR image feature information is introduced, and an image-representation-constrained reconstruction model is proposed. Through simulation experiments, it is verified that this SAR deep unrolling network imaging method can efficiently and accurately perform undersampled SAR imaging and has strong generalization ability for different imaging scenes and radar parameters. The following specific conclusions are drawn from the research: the NSR-NET uses the general image representation in SAR images, such as sparse features and edge textures, to achieve fast and accurate undersampled SAR imaging in nonsparse scenes. Under the lowest undersampling rate of 30%, the structural similarity index is improved to over 0.9, which is more than 0.4 higher than the comparison method. Moreover, this method has good adaptability to different imaging scenes and radar parameters.

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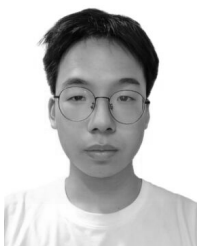

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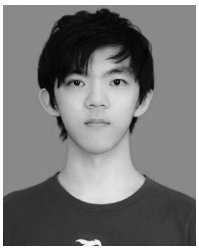

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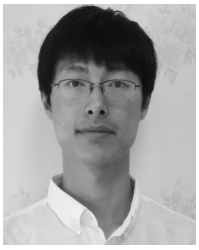

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